What if everyone follows the same best advice?

A new road can worsen traffic equilibrium.
Can more accurate shared advice do the same when people compete for limited opportunities?

Shared Intelligence, Competing Decisions

A mathematical working paper • Synthetic evidence • Not peer reviewed

THE RESEARCH PATH

Four papers sharpen the question

1968 / 2005	**Braess's traffic paradox**	A new route can worsen equilibrium travel times.
2018	**Informational Braess' paradox**	Learning about routes can hurt affected travelers.
2021	**Algorithmic monoculture**	A more accurate common ranking can reduce collective hiring performance.
2025 / 2026	**Strategic hiring**	Congestion-aware differentiation changes the outcome.

Our question: when does the private gain from accuracy create a public cost?

THE MODEL

Accuracy is only one part of payoff

100 participants • 2 opportunities • Congestion cost per competitor

$$E[W] / N = \mu - c(N - 1)q$$

μ Expected intrinsic value

Better advice raises the chance of selecting the good opportunity.

q Probability two actions coincide

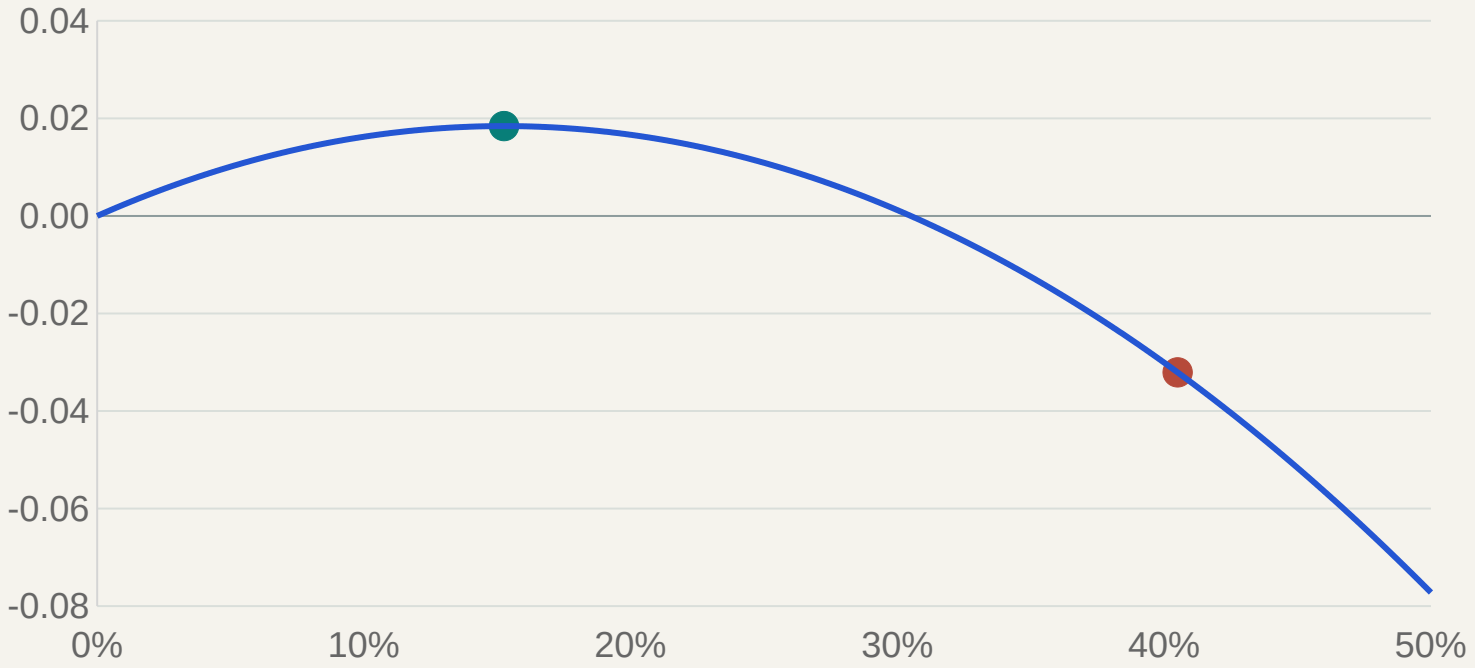
Shared advice raises crowding when it sends people to the same place.

Private accuracy: 60% • Shared accuracy: 80% • Payoff = value − congestion

THE RESULT

Voluntary adoption overshoots the optimum

Payoff change from private-only baseline



15.3%

Payoff-maximizing independent adoption

40.5%

Equilibrium adoption
Loss: 0.03208 per person

Exact model calculation. The loss is about 0.35% of baseline payoff, not a measured economic loss.

THE REVIEW → THE REFINEMENT

The result can survive after signals arrive

Each person sees only their own private and shared recommendations.

Illustrative calibration; opponents use the equilibrium strategy.

WHEN THEY AGREE

Follow the recommendation.

Gain from following: 401 / 875 > 0

WHEN THEY DISAGREE

Either action is optimal.

Mixing at p = 40.5% sustains equilibrium.

A scoped equilibrium result. No claim of real-world losses, chaos, or unique equilibrium.

github.com/triggeredcode/shared-intelligence-competing-decisions